Free-Tier LLM Gateways as AI-Text Detectors

A dated benchmark of model accuracy, quota regimes and out-of-sample stability across four providers, with a band-scoring protocol for production use

Preprint / technical report — not peer reviewed. This file is the two-part edition: Part I (§1–§8) plus Part II (§9).
Citation: deposited at doi.org/10.5281/zenodo.23460013.
doi.org/10.5281/zenodo.23460012 is the concept DOI and resolves to the newest version of the record.
Part I (§1–§8) measurement window: 2026-09-28 to 2026-10-13, published as it was measured.
Part II (§9) measurement window: 2026-10-13 to 2026-10-17 — it extends Part I with the length-band, evidence-output, de-formatting and deployment-regime measurements, and it corrects the claims later evidence refuted.
Reading note: where Part II supersedes a Part I claim, the Part I text is kept as the dated record and carries a † marker pointing into §9. Every availability, quota and latency claim in both parts is time-bound to the window that produced it.
A printable copy is at report.pdf.
License: CC BY 4.0 (text). Model names and trademarks belong to their providers. Corpus content is third-party data and is not redistributed here (see §8, Data availability).

Abstract. General-purpose LLMs asked to output a structured AI-ness verdict can serve as text detectors at near-zero cost, and their availability is governed less by model quality than by the free-tier policies of the gateways that serve them. This report benchmarks 20+ models across four gateways (OpenRouter free pool, NVIDIA NIM free tier, Google AI Studio free tier, and a local multi-account router) on a 48-text in-sample corpus and a 48-text disjoint holdout drawn from HC3, all scored through the same strict-JSON prompt, a fixed decision band (AI ≥ 0.69, HUMAN ≤ 0.40) and a confidence gate (≥ 0.60). Three findings stand out. First, *model choice dominates tuning*: identical prompts and parsers span 0.31 to 1.00 corpus accuracy, and one free model (`moonshotai/kimi-k3`) was the only configuration measured with zero wrong decisions on both corpora, reproduced in a same-day repeat run (AUC 1.000 both captures; McNemar vs. incumbent lanes p = 1.0000). Second, each gateway enforces a materially different regime — OpenRouter has two 429 classes that require opposite remedies, Google's per-model daily quota (nominally 20) *delivers* a volume that is per model *and* per project (5–46 requests in one window observed) and stacks linearly with extra project keys, and NVIDIA free serving windows open and close within hours — so a free model's measured accuracy and its usability as a live lane are separate, per-gateway questions. Third, in-sample perfection does not transfer: a lane that erred zero times on the corpus its band was fitted on produced a false negative out of sample, while the one lane that reproduced zero-wrong out of sample deferred 2–3 of

48 texts per run. A practitioner's decision table and an explicit limitations section accompany the results.

Part II adds four findings and one retraction. (1) *Accuracy does not survive listing*. The lane that Part I ranked first was delisted by its gateway four days after the Part I window closed (HTTP 410, permanent), and another advertised model answers 404 on every real call while still appearing in the catalogue — so a detector's availability must be probed per model, per gateway, on the day of use, and `/v1/models` is not a health check. (2) *Text length is a first-order variable*: on 48 texts of ≥3000 characters the shipped band made zero wrong calls, while on 24 texts under 300 characters the same rule made 6 wrong calls out of 24; the band refit for that regime erases them but cuts coverage from 92% to 29%. (3) *Surface formatting was inflating the headline*: removing scrape noise from the human side moved the two-model ensemble from 0.938 to 0.875 acc/corpus and produced its first AI-passes-as-human miss, while AUC barely moved — the ranking metric was blind to it. (4) *Detectors can be made to show their work*: asking for verbatim evidence spans in the same strict-JSON reply yields quotes on most answered texts (575 over 144), and their strengths occupy 0.30–0.95, not 0–1, which changed how the product renders them. The retraction: the two-model tie-break that Part I replayed at 0.938 decided 3 of 48 texts on the shipped primary and got all 3 wrong, so it now ships disabled; on the evidence measured here, a single calibrated lane beats an unvalidated second opinion.

1. Introduction

Commercial AI-text detectors are calibrated to the models that existed when they were trained, and they are expensive per request. An alternative used here: ask a general-purpose LLM, through a plain chat API, to score a text and emit a structured verdict, then apply fixed decision thresholds. Because the cheapest LLM access today is free-tier gateway capacity, the practical question for anyone building a low-cost verifier splits in two: *which models score well*, and *which serving channels stay reachable*. This report measures both halves on the same protocol, dated, with every availability claim tied to the gateway and window where it was observed.

The work was performed as engineering validation of a web verification app, and this report is written to be usable as its published evidence base: §2 defines the protocol, §3–§5 give results, §6 is a decision table for choosing a free detector model, §7 states the limitations without softening them. §9 is Part II: the same protocol continued to 2026-10-17, after the detector became a deployed product.

2. Method

2.1 Corpora

Both corpora are drawn from the Human-vs-AI Generated Comparison corpus (HC3) [1], which pairs real user posts with ChatGPT-3.5-era answers across six domains (medical Q&A, Reddit ELI5, open QA, wikiCSAI, finance, law).

- **In-sample corpus** — 48 texts (24 AI / 24 HUMAN), ~85-word human median.
- **Holdout corpus** — a disjoint 48-text two-sided set (24 AI / 24 HUMAN) taken from a non-overlapping offset window of the same source; disjointness was verified by normalized text overlap and by question overlap (0 of 48 shared), and independently re-checked against the source with a negative control planted on a tampered copy.

A third control corpus of 46 real book excerpts (Project Gutenberg, all-human) was used to test false-positive behaviour and is reported only as a caution: it is one-sided and saturated every model at specificity 1.000, so it cannot rank detectors (see §7).

2.2 Scoring protocol

Every model received the same single-turn prompt requesting a strict JSON object with `ai_probability` (0–1), `confidence` (0–1) and short signals, at temperature 0.1. Notable protocol choices, each learned from failure:

- **No `response_format`.** On free tiers, JSON-mode requests caused HTTP 400s, reasoning preamble leakage and malformed output; it was removed and replaced by a defensive parser (markdown-fence strip → balanced-brace scan → regex salvage). Salvaged rows are always flagged and excluded from headline numbers.
- **Transport failures are separated from model failures.** Each capture records no-answer vs bad-JSON vs HTTP-error vs timeout per row; a coverage gap attributable to a gateway is never quoted as model evidence without its *n*.
- **Every "unreachable" finding names the gateway and date.** The same weights proved reachable on one provider and permanently dead on another (§5).

2.3 Decision band and metrics

Verdicts follow a fixed band: **AI if p ≥ 0.69**, **HUMAN if p ≤ 0.40**, otherwise deferred (review); a deferral also results when the lane reports confidence < 0.60. Metrics:

- `acc/corpus` = correct ÷ texts given (deferral costs accuracy) — the deployment-relevant column;
- `acc/answered` = correct ÷ texts the model actually decided — the judgement-quality column;
- AUC via Mann-Whitney over answered rows (ties 0.5) — ranking quality, threshold-free;
- Paired significance via exact McNemar on decisive-vs-decisive disagreements [2], joining the two captures by text id and applying one rule to both sides.

The band was calibrated on the in-sample corpus, which makes the holdout runs the only unbiased numbers in this report; per-model bands (fitted cuts, two-fold) were also computed for several lanes and are reported where they change the story.

2.4 Gateways under test

Gateway	Access path	Class

OpenRouter	15 `:free` model ids; up to 6 free accounts (class-(b) headroom stacks linearly across accounts)	Free
NVIDIA NIM	free-tier endpoints; 80-model catalogue, ~20 genuinely servable on one account	Free
Google AI Studio	2 working free accounts + 1 rejected at auth (per-model buckets are per project, so each working account adds capacity), ~62 model ids	Free
Local multi-account router	a self-hosted OpenAI-compatible router holding OAuth and key-based upstream accounts	Free + paid
Paid reference pair	`nemotron-3-ultra-550b` primary + `diffusiongemma-26b` tie-break	Paid baseline

3. In-sample results (48-text corpus)

All rows answered 48/48 unless *n* says otherwise; band as in §2.3. Two accuracy columns are given because they answer different questions (see §2.3); only `acc/corpus` is comparable across coverage.

Model	Gateway	n	AUC	wrong	deferred	acc/corpus	med AI	med HUMAN
kimi-k3	NVIDIA free	48	1.000	0	3	0.938	0.860	0.220
glm-5.3	NVIDIA free	48	1.000	0	3	0.938	0.930	0.135
glm-5.3-flash	NVIDIA free	48	1.000	1	2	0.938	0.930	0.210
claude-sonnet-4.5 (router/OAuth)	router	48	0.988	2	0	0.958	0.850	0.150
claude-sonnet-4.5 no-think (router)	router	48	0.986	2	0	0.958	0.850	0.150
ling-3.1-flash	OpenRouter free	48	0.986	1	12	0.729	0.850	0.325
gemini-3.1-flash-lite	AI Studio free	46	0.955	13	0	0.688	0.920	0.835
gemma-4-31b-it	NVIDIA free	48	0.945	7	0	0.854	0.980	0.150

paid 2-model pair (tie-break replay)	NVIDIA paid	48	—	3	0	0.938	—	—
nemotron-3-ultra-550b (paid primary alone)	NVIDIA paid	48	—	2	10	0.750	0.815	0.250
openrouter/free (router model)	OpenRouter free	33	0.743	7	7	0.396	0.850	0.680
lfm-2.5-2.6b:free	OpenRouter free	47	0.480	16	16	0.313	0.665	0.650
llama-3.2-11b-vision-instruct	NVIDIA free	48	≈0.0	32	0	0.333	0.200	0.700

Observations that survived the whole measurement window:

- **Model choice dominates tuning.** The same prompt and parser span 0.313 → 0.938 acc/corpus; temperature/top_p sweeps move a model far less than switching models does.
- **AUC cannot see a band problem.** Several lanes ranked classes perfectly while flagging most humans as AI under the shipped cuts (a router route at AUC 1.000 with human median 0.72; a Google lite lane at AUC 0.955 with human median 0.835 and 13 of 24 humans misclassified, §5.2). Only band-scoring exposes this; ranking alone is not evidence of deployability.
- **Detectors can be inverted.** A free vision model scored human text as *more* AI-like than AI text (separation −0.500). Any unmeasured model is a liability; this is the case for labelling every detector choice out of sample.
- **Run-to-run jitter is a pipeline property.** Three runs of one identical model scored 0.458 / 0.404 / 0.378 on the same corpus; on a borderline screen subset, 8 of 12 texts flipped verdicts between identical reruns. Single-text or single-run judgements are meaningless; aggregates and paired tests are the only reliable comparisons.

4. Out-of-sample results (disjoint 48-text holdout)

Lane	Gateway	n	AUC	wrong	deferred	acc/corpus	decisive miss
kimi-k3 (run 1)	NVIDIA free	48	1.000	0	2	0.958	none
kimi-k3 (run 2, same night)	NVIDIA free	48	1.000	0	3	0.938	none

claude-sonnet-4.5	router/OAuth	48	—	1	0	0.979	hu-007 → AI (p 0.72)
glm-5.3	NVIDIA free	48	—	1	2	0.938	ai-012 → HUMAN (p 0.35)

- **In-sample perfection transferred for one lane and failed for another.** `glm-5.3` had erred zero times in sample with every AI text above 0.72; on the holdout its AI floor collapsed to 0.35 — the celebrated 0-wrong property was a fact about those 48 texts. `kimi-k3` kept zero wrong on the holdout *and* reproduced it in an independent same-night rerun.
- **The recurring adversarial text.** One holdout AI text (`ai-012`) defeated every other configuration that met it (glm-5.3: 0.35; the paid ensemble on cleaned text: an AI→HUMAN miss at 0.38). kimi-k3 decided it correctly as AI in both runs (0.78 / 0.74, confidence ≥ 0.6 both). One text is not proof, but it is the second occurrence of the same failure class — false negatives concentrate, and measuring them needs a two-sided holdout, not a bigger pile of human text.
- **Jitter direction on kimi-k3 was benign.** Between the two identical-parameter runs, 33 of 48 scores drifted (±0.04–0.10 typical) and exactly 3 verdicts moved — one defer became a correct AI call, two correct HUMAN calls became defers — *none moved into a wrong decision*. On this lane, jitter buys review volume, not error.
- **Pairs differ by too little to claim.** McNemar exact tests on all 48 shared texts: kimi vs sonnet, one discordant pair (kimi right); kimi vs glm-5.3, one discordant pair (kimi right); both p = 1.0000. At n = 48 a one-error advantage is not statistically separable — the defensible claim is structural (the only zero-wrong-on-both-corpora, reproduced), not statistical dominance.

5. Gateway regimes: availability is per-provider, dated, and decisive

5.1 OpenRouter free pool — two 429 classes with opposite remedies

- *Class (a), upstream per-model shared pool*: transient, can last indefinitely, **unfixable by adding accounts** — one model returned 429 on all 6 accounts × all retries in three independent confirmations spanning hours (0/12, 72 requests, nothing recovered).
- *Class (b), per-account daily cap*: printed inside the 429 body (visible only on failure; successful responses carry no rate headers); **stacks linearly across accounts** — a 6-account rotation recovered 32 missing capture rows across two quota windows with zero harness-side failures. The visible cap number varies by account history (50/day measured on no-credit accounts, up to 200/day where credits exist).
- A **404 is terminal**: once a free id is delisted its lane freezes permanently at whatever coverage it had; AUC must always be quoted with *n*.

- Gateway errors also arrive *inside* HTTP 200 bodies; harnesses that only retry on transport status misattribute these as model failures and understate the model.

5.2 Google AI Studio free tier — per-model buckets whose delivered volume is per-lane and per-project

- Quota metric is *per project, per model, per day*, stated at 20 requests; actual **delivered** volume varied enormously by lane and project: preview-family lanes delivered ~5–6 rows/day on project 1 and 16 rows in one window on project 2, while a lite-family lane delivered **46 of 48 rows in a single window**. Sleeping past the server's own `retryDelay` bought exactly one row, not a bucket — the printed delay is not a refill signal. Preview model names bill into the base model's bucket; `-latest` aliases into another.
- Because the bucket is per project, **additional project keys stack capacity linearly** — the same property OpenRouter's account rotation shows — and each key must still be probed per model: a third key that Google rejects with `400 Invalid Auth key` contributes nothing.
- Two independent limiters: a per-model bucket can be exhausted while the model still returns "high demand" 503s (these *are* transient), so capacity noise and quota starvation coexist.
- Consequence for throughput planning: **delivered requests/day/model is a measured per-lane fact, not a gateway constant** — any production detector on this tier must quota-probe its own exact model on its own project before sizing. On this gateway quota, not accuracy, was the binding constraint even though its best screen (a flash-family model) was the strongest measured anywhere: AUC 1.000, 0 wrong, human median 0.06 on 12 texts; its 21-row partial capture at window close still reported AUC 1.000 with 0 wrong.
- Consequence for validation, from the high-throughput lite lane: it delivered a *full corpus in one window* and was still band-dead — AUC 0.955 with **13 wrong calls, every one a human text read as AI** (human class bimodal, median 0.835, upper mode overlapping the AI class). High free throughput is a reason to *measure* a lane faster, never evidence that the lane is a detector.

5.3 NVIDIA NIM free tier — serving eras, not service levels

- The catalogue endpoint lists 80 models; 55 return "function not found" on real calls. */v1/models is a promise list, not an entitlement list.*
- Models *revive and die in hours*: one opened a ~12-hour serving era (12/12 screen, AUC 1.000, the lowest human median ever measured at 0.075), collapsed to 2/48 overnight and answered nothing the next morning — an era boundary, not a time-of-day effect. Probe before spending a capture; a 2-text gate costs almost nothing.
- Latency is pool load, never a model property: p50 for the same lane moved 13.6 s → 96 s between windows on the same day; a full corpus took 144 s once and 5183 s another time, both with zero failures.
- Hard parameter constraints exist and are discoverable only in the 400 body (e.g. one model fixes top_p at 0.95 and rejects any override). *A validation 400 is a harness*

constraint, not a model verdict.

5.4 Local multi-account router

- Aggregating many upstream accounts (OAuth and keys) behind one OpenAI-compatible endpoint proved the **most reliable capture channel measured**: two full-corpus runs, 48/48 both, zero salvage/truncation/errors.
- Its virtual "auto" routes rank well but band badly (§3) — a router's routing objective is not a detector's decision objective.
- Free OAuth tiers behind such routers carry provider-side risk (session expiry, policy change) that a direct key does not; treat them as convenience channels, not SLOs.

6. Which free model to use today (decision table)

Conditions of use: October 2026, short-form English prose (~85–200 words), a strict-JSON single-prompt protocol, and the band in §2.3. Re-measure before deploying: every row here is a dated measurement, not a subscription.

Rank	Lane	Gateway	acc/corpus in-sample	acc/corpus holdout	approx. median latency (s)	main constraint
1†	kimi-k3	NVIDIA free	0.938	0.958 (reproduced)	14	top_p must be 0.95; serving eras can end
2	claude-sonnet-4.5 via OAuth router	router	0.958	0.979	3	OAuth session, not a key; 2 wrong (both human text)
3	glm-5.3	NVIDIA free	0.938	0.938	22	one AI→HUMAN miss on holdout; pool intermittently silent
4	ling-3.1-flash	OpenRouter free	0.729	not yet measured	—	12/48 deferred; best free-OR rank (AUC 0.986)

5‡	glm-5.3-flash	NVIDIA free	0.938	not yet measured	8	fastest strong lane; holdout pending
—	Google flash-family (3.8-flash)	AI Studio free	partial: 21/48, AUC 1.000, 0 wrong	—	18	delivered volume is per model *and* per project; stack project keys, quota-probe before sizing

- † **Part II correction (2026-10-15): this row is void on its gateway.** Four days after the Part I window closed, `moonshotai/kimi-k3` began answering HTTP **410 Gone** on NVIDIA NIM (first probe: 410 in 91 ms). 410 is a fourth NVIDIA failure class beyond 429/503/400 and, unlike those three, it is *permanent* — a delisted lane freezes at whatever coverage it had. Its zero-wrong measurements stand as a record of that window; its candidacy on this gateway does not (§9.1).
- ‡ **Part II: rank 5 is the lane the product actually ships.** It is the only candidate that later answered 48/48 on a long-text corpus, 24/24 on a short-text corpus and 42/42 on a 3000–6000 character stress corpus with zero truncation, and it has since been carried end-to-end through the live webhook stack — the property §6's own caveat said no lane had (§9.3, §9.4, §9.8).
- **kimi-k3's holdout row above is its first run.** The same-night rerun also had 0 wrong but deferred 3 texts instead of 2 (acc/corpus 0.938): what reproduces exactly is the absence of errors; the deferral count is what jitter moves.
- No free lane here has been verified end-to-end through a live webhook at time of writing[§]; the paid two-model pair is the only configuration ever checked through production plumbing, and its live score (41/48) was *below* its offline replay (45/48) — a structural gap, because a secondary calibrated on one primary run is untested against a drifting primary run. Deploy-time evidence must be collected through the real stack.
- Ranking two lanes is free-tier-fragile by nature. A resilient low-cost setup runs the best lane as primary with a *different gateway*'s second-best as fallback, because cross-provider failure modes are independent while same-provider ones correlate.

[§] Closed in Part II: a free lane (`z-ai/glm-5.3-flash`) now scores real documents through the whole deployed path — browser → reverse proxy → gunicorn → n8n → provider → parser → verdict — and twelve end-to-end assertions pass on it, including two labelled texts answered with phrase evidence attached (§9.8).

7. Limitations

- **Small, single-source corpora.** 48 + 48 texts, both from HC3. The AI class is ChatGPT-3.5-era; transfer to modern frontier-model output is untested here and is the biggest open risk.

- **Verdicts are style proxies, not truth.** An independent commercial detector (GPTZero) run on the same corpora flagged de-formatted *fresh* AI text at 100% while reading genuine ChatGPT-era HC3 answers as 96% human — both instruments key on surface style and era-calibrated signal, not on origin. Absolute verdicts on out-of-era text from any detector here, LLM-based included, should be treated as probabilistic.
- **Length confound.** AI texts average ~2× the words of human texts in both corpora; models may exploit length rather than style. A length-matched rerun is designed but not yet executed. Likewise the band was fitted on ~85-word text; its behaviour on 500-word documents is unmeasured. *Part II status:* the length-*band* half of this is now measured (§9.3) — accuracy holds at ≥3000 characters and collapses under 300 — but the matched-pair rerun that separates "long" from "stylistically different" is still not executed.
- **Band transfer risk.** The 0.69/0.40 cuts were fitted in sample; several lanes have measurably different optimal cuts, and applying one band across models biases weaker-but-calibrated rows. Per-model bands were stored for a subset; comparisons in §4 deliberately used one rule for both sides. *Part II status:* this became the shipping rule — every stored model row carries its own two cuts and the ranking table scores each row at its own band (§9.8), and §9.7 adds why a cut must not sit on a value the model actually emits.
- **All availability, quota and latency claims are dated.** Free tiers change without notice; the catalogue itself lies in both directions (listed models dead; unlisted weights served). Every number here says which gateway and which window it came from, and none of it should be quoted as a property of a model alone.
- **Not peer reviewed.** Single-lab (single-engineer) measurements, no held-out implementation verification by a third party.

8. Reproducibility and data availability

- Corpora are rebuildable from the public HC3 dataset with the recorded offset windows [1]; per-document provenance (url, retrieved date, char offsets, sha256) is kept for every row. Raw text is not redistributed.
- Every capture is stored as append-only per-row JSONL (id, label, p, confidence, signals, latency, error class, attempt) with a run manifest recording every parameter of the command that produced it. The scoring and paired-test code (band scorer, Mann-Whitney, exact McNemar, corpus disjointness verifier with negative control) are plain JS, <200 lines each.
- The prompt, parser rules and band constants are exactly those listed in §2. The per-model reproduction checklist is not behind a request: the record at the DOI above ships `model-ranking.csv` (one row per measured model, in the order the application ranks them), `per-text-scores.csv` (3,937 rows of label, probability, confidence, answered flag and error class — one per run and text, no corpus text redistributed), `runs-manifest.csv` (144 rows recording the parameters of every capture) and a `README.md` that defines each column. Re-scoring `per-text-scores.csv` at each model's own exported band reproduces the correct / wrong / decided / accuracy figures of all 48 ranking rows, so every number in this report can be traced from the dataset to the command that produced it.

9. Part II — 2026-10-13 to 2026-10-17: what a deployed system measures that a benchmark cannot

Part I stopped at the boundary of the measurement harness. The four days after it were spent putting the same protocol behind a real web app (reverse proxy → WSGI server → workflow engine → provider lane) and shipping the result, which produced three kinds of evidence a benchmark cannot: an availability failure that arrived *after* a lane had been recommended, two measurement axes nobody had varied (text length, and surface formatting of the human class), and a detector output field that turns a verdict into shown evidence. The prompt, parser, scoring rule, append-only JSONL capture format and run-manifest discipline are unchanged from §2; where a number below is at a model's *own* stored band rather than the §2.3 fixed band, it says so.

9.1 Availability after the window: the recommended lane died, and a listed model never lived

- **HTTP 410 is a fourth failure class, and the only permanent one.** `moonshotai/kimi-k3` — Part I's rank 1, zero wrong on three corpus families — began answering **410** on NVIDIA NIM on 2026-10-15 (first probe: 410 in **91 ms**, body: "reached its end of life on 2026-10-15T08:00:00Z"). Unlike 429, 503 and the 400-class parameter rejections, none of which are permanent, this one cannot be waited out: the in-progress long-text capture froze permanently at **39 usable of 48** rows, and the band-patching tool now refuses to write that model's row at all — its measured band is kept as a record, not treated as a lane. A detector ranking is therefore a *dated* document in a stronger sense than §7 allowed: the recommendation can be voided by an upstream decision that produces no warning in the API.
- **A catalogue entry is not a health check.** `GET /v1/models` on the same gateway answered 200 with **81 ids** and `nvidia/nemotron-3-ultra-550b-a55b` was exactly one of them, while every chat completion to it returned **404** "function not found" (first probe: 404 in 385 ms) — reproduced on two independent keys and relayed identically by the OpenRouter `:free` mirror of the same weights, i.e. the same function id fails on both paths. §5.3's "the listing cannot be used as an entitlement list" generalises: *it cannot be used as a liveness check either*, in either direction. The shipped default had to move to a fallback lane because of it.
- **Method consequence.** A two-text probe (8 `max_tokens`, so it costs almost nothing) now precedes any capture, and the same probe is what the product's admin panel reports. Probing is not optional hygiene: on this gateway, listing and serving are decoupled.

9.2 The transport ceiling sets the longest document you can score

NVIDIA free serving imposes a **~300 s server-side ceiling** (measured 2026-10-14: `integrate.api.nvidia.com/v1` returns HTTP **504 at 302 s** regardless of the client's own timeout; one model 504'd even with a 450 s client timeout). No client-side setting outruns it, so the practical maximum document size is set by the gateway, not by the application. A short-

prompt probe of eight rows on one day shows how wide the spread is between "serves" and "usable":

Model (row)	Status	Wall time	Reading
nemotron-3-ultra-550b-a55b (1, then-deployed primary)	404	0.4 s	listed but not deployed — dead
diffusiongemma-26b-a4b-it (2, tie-break)	200	1.4 s	healthy
glm-5.3 (22)	200	269 s	answers, but 4.5 min for 16 tokens
gemma-4-31b-it (21)	504	302 s	gateway ceiling, no verdict obtainable
glm-5.3-flash (23)	200	80 s	slow, usable
nemotron-3.5-lightning-30b-a3b (24)	200	134 s	slow, usable
deepseek-v4.1-flash (25)	200	301 s	answers at the gateway's own ceiling
kimi-k3 (26)	410	0.09 s	end-of-listing, permanent

Two products of this: the application must distinguish *slow* from *absent* — Windows hangs rather than refusing on a dead local port, so a timeout and an unreachable engine used to produce one identical message; it now discriminates them with one bare TCP handshake after the failure and reports `ENGINE_TIMEOUT` vs `ENGINE_UNREACHABLE` under the same 502 contract. And a single timeout number is a whole chain: lane ceiling → caller wait → WSGI worker timeout → proxy read timeout, each tier above the one inside it (§9.8).

9.3 Text length is a first-order variable, not a confound to be matched later

Two new corpora were built from the same source rows, by length only: **`corpus-long` , 48 texts ≥3000 characters**, and **`corpus-short` , 24 texts <300 characters**. `glm-5.3-flash` answered 48/48 and 24/24 with zero truncated and zero salvaged rows. Scored twice — once at the Part I fixed band (0.69/0.40, confidence gate 0.6), once at the model's own saved fitted band:

Corpus / lane	Band	decided	wrong	deferred
long 48 — glm-5.3-flash	0.69/0.40/gate 0.6	48/48	0	0
long 48 — glm-5.3-flash	saved 0.90/0.35/gate 0.65	44/48	0	4
long 39 — kimi-k3 (frozen capture)	0.69/0.40/gate 0.6	39/39	0	0

short 24 — glm-5.3-flash	0.69/0.40/gate 0	22/24	6 (2 FP, 4 FN)	2
short 24 — glm-5.3-flash	0.69/0.40/gate 0.6	14/24	2	10
short 24 — glm-5.3-flash	saved 0.90/0.35/gate 0.65	7/24	0	17

- **Long text is comfortable for every band tested.** Length pushes probabilities to the extremes, so a cut that is merely adequate at 800 characters is at ease at 3500. The refits neither break long-band accuracy nor improve it (flash 48→44 decided, wrong count unchanged) — which means the long/3–6k captures must *not* be cited as evidence *for* the refitted cuts. On this axis they are equivalent.
- **Short text is where a band bleeds.** The Part I band put out 6 wrong calls on 24 fragments; the model's own saved band erases them (0 wrong) at the price of coverage 92% → 29%. On a 200-character fragment there is little style to judge, so deferring is arguably the correct verdict rather than a failure — and §7's length warning now has a number attached to it.

9.4 A harder AI class separates models that the HC3 class cannot

A stress corpus of 42 rows at 3000–6000 characters (33 HUMAN from verifiable published sources, 9 AI) was scored at each lane's own saved band:

Lane	Band (AI/HUMAN/gate)	decided	wrong	AI score range	HUMAN score range
claude-sonnet-4.5 (router)	0.75/0.30/0	42/42	0	0.85–0.92	0.02–0.15
glm-5.3-flash (NVIDIA free)	0.80/0.55/0.50	42/42	0	0.88–0.95	0.01–0.05
glm-5.3 (NVIDIA free)	own saved band	37/37	0	—	—
nemotron-3-ultra-550b (paid flagship)	0.69/0.40/0.60	39/41	5	—	—

Five of the nine stress AI articles scored 0.15–0.30 on the paid flagship *with confidence 0.75–0.88* — confidently wrong HUMAN calls on the same texts every other model reads at ≥0.85. A grid sweep over old-fit data plus stress jointly found no band that rescues it: the only zero-stress-error cuts (e.g. 0.80/0.05) decide just 24 of 96 old rows and 34 of 41 stress rows, i.e. they avoid the error by abstaining on almost everything. That is a capability gap on this AI class, not a threshold misfit. Combined over *all* corpora at saved bands: sonnet 272 decided / 2 wrong, glm-5.3-flash 195 / 3, glm-5.3 172 / 0, nemotron-3-ultra 176 / 11. Provenance caveat: the stress AI

class is self-generated, so it is weaker evidence than the HC3 rows, whose provenance is a public dataset.

9.5 The human class was partly being detected by its formatting noise

The HC3 human rows carry scrape artefacts (stray whitespace, broken bullets, escaped characters). A cleaned copy of both corpora was built and every lane re-captured on it. AUC barely moved; band-scoring and effect size did:

Lane	AUC dirty → clean	Cohen's *d* dirty → clean	human-class-as-AI count dirty → clean
claude-sonnet-4.5 (corpus)	0.988 → 0.987	5.017 → 4.265	2 → 3
nemotron-3-ultra-550b	0.934 → 0.935	2.444 → 2.246	2 → 3
diffusiongemma-26b	0.954 → 0.935	1.729 → 1.843	12 → 10
glm-5.3-flash (free)	1.000 → 0.997	5.884 → 3.967	1 → 2
openrouter ultra:free	0.937 → 0.906	—	0 → 1
MiniMax-M2.5 (free)	0.802 → 0.882	—	14 → 10

The two-model ensemble in Part I's deployment configuration went from **45 correct / 3 wrong / 0 review, acc/corpus 0.938** on dirty text to **42 / 5 / 1, acc/corpus 0.875** on clean text — and gained its first *AI-passes-as-human* miss (`ai-021` , created by the tie-break rule). The sign is model-specific: one free lane's AUC *improved* 0.802 → 0.882. The reading we publish is the conservative one: **0.938 should be treated as an upper bound inflated by formatting noise; ~0.875 is what the same thresholds earned on clean in-sample text.** A detector benchmarked only on dirty text is partly measuring the scraper.

9.6 Evidence in the reply: verbatim spans, and what their strengths are actually worth

The prompt was extended to request, beside the verdict, up to four *verbatim* spans with a direction (AI or HUMAN) and a strength 0–1 — so the product can paint the evidence rather than assert a number. Across three full captures this yielded **575 quotes over 144 answered texts** (192 + 192 + 191 rows), **355 AI / 220 HUMAN**. Structural facts that mattered:

- **A truncated answer loses the evidence too.** Rows flagged salvaged or truncated carry *no* quotes and no signals — the same capture-degradation rule §2.2 applies to `ai_probability` applies here, so evidence counts must be quoted with their *n*.
- **Strengths are not spread over 0–1.** The observed domain is **0.30 – 0.95** (44 distinct values overall, 40 among AI quotes), with AI p25 0.71 / p50 0.80 / p90 0.88 and HUMAN p50 0.70 / max 0.92. Three quarters of AI quotes sit in 0.71–0.95 — the crowded top of the

scale. The UI ramp was therefore rescaled onto the measured domain instead of 0–1: across the interval that actually gets used (AI p25 → p90) the green channel now spends **58 units where it spent 17**, hue separation roughly doubles exactly where the data is crowded, while the alpha *ceiling drops* 0.699 → **0.600** and the palest quote renders at 0.26 instead of 0.426. The previous shape painted the weakest evidence any lane ever reported (0.30) as deep, certain orange — a design lie made visible by measuring the domain.

- **Are the quotes really verbatim?** Located all 575 against their source texts: exact substring **574 (99.8%)**, whitespace-normalised fallback 0, plus one case-insensitive fold → **575 / 575 located**. A punctuation-insensitive step was measured and rejected: it recovered nothing (0), so it was not shipped. Live in the panel, the failure mode is reported rather than hidden — one detect returned 3 quotes and 1 could not be found verbatim in the input, and the panel said so.
- **Limitation of the ramp:** its stops describe *the deployed primary's* output habits. A different primary would need the domain re-measured, not the constants reused.

9.7 Removing the generation cap: a paired control, and bands that had to be re-earned

Every capture had been produced with `max_tokens` capped (700–1400). Because a cap can truncate the JSON the parser then has to salvage, the cap was removed entirely (`max_tokens = 0` ⇒ the field is omitted), the "keep the response under 400 tokens" prompt tail was deleted from both copies of the prompt (asserted byte-identical, 785 characters each), and all 31 stored model rows were set to uncapped. That made the saved bands measurements of *a configuration that no longer existed*, so they were re-validated at the same criteria over the two 48-text corpora joined (92 scorable texts):

Lane	Band	decided	wrong	acc/decided	coverage
claude-sonnet-4.5	0.80/0.30/0	86/92	0	100%	93%
claude-haiku-4.5	0.75/0.35/0	49/92	0	100%	53%
diffusiongemma-26b	0.90/0.30/0	76/92	7	90.8%	83%
nemotron-3-ultra-550b	0.69/0.40/0.60	not re-validatable — the model is unreachable at the provider			

- **The cap was not the variable.** Exact McNemar on the capped vs uncapped captures of the same 48 texts: **p = 1.0000, 0 discordant pairs**, mean |Δp| 0.0115. No decision changed.
- **But it moved the emitted lattice, and cuts live near it.** These models emit a handful of distinct probabilities, not a continuum. Uncapping added a value (0.72) between the AI cut and a text that had scored 0.75 — turning a false positive into a deferral. The corollary is a fitting rule the earlier grid missed: a cut placed *on* a populated value makes that text's verdict hinge on rounding, so band fitting now tie-breaks on lattice margin (haiku's margin

0 exposed this: its cut sat exactly on values 0.72/0.75/0.78 it emits). Explicit exchange rate at the chosen operating point: about **4 deferrals bought per false positive removed**.

- **Not every lane has a good uncapped answer.** Diffusiongemma's 7 errors are all human-as-AI at 0.92–0.98 probability with 0.92–0.98 confidence, while the grid's highest candidate cut is 0.90 / gate 0.75 — *no band in the search space can exclude a mistake the model is sure of*. The best the grid offered was one extra decision (77 vs 76) for the same 7 errors; it was not written to the database, because it does not meet the 95% accuracy floor. Seven errors is that lane's floor, not a misfit.

9.8 What actually ships now, and the evidence for each choice

- **Single calibrated lane as the deployed default:** `z-ai/glm-5.3-flash` at its own stored band (AI ≥ 0.80, HUMAN ≤ 0.55, confidence floors 0.50). Knee refit over train+val: **156 decided, 3 wrong, acc/decided 0.981**; 42/42 correct on the stress corpus (§9.4); 195 decided / 3 wrong across all corpora. Caveat that stays open: that band was fitted on 2026-10-15, *under the token cap*, so its uncapped re-validation is owed even though the cap control in §9.7 found no verdict change.
- **The second-opinion lane now ships switched off.** Faithful replay of the deployed rule over the 48-text clean corpus (primary as above, diffusiongemma tie-break at 0.90):

Rule	correct	wrong	deferred	acc/answered	answered
shipped rule: secondary decides every deferral	44	4	0	0.917	48
secondary may not create an AI verdict over a human-leaning primary	44	2	2	0.957	46
secondary lane off	44	1	3	0.978	45

  The tie-break decided 3 of 48 texts and got *all 3 wrong* (`hu-004` primary 0.78; `hu-009` 0.50 conf 0.40; `hu-019` 0.45 conf 0.35 — three human texts, each answered AI by a secondary reading 0.95–0.98). This is a property of *this* primary, not a verdict on the lane: replayed against the previous primary with its own ruler, the tie-break decided 15 of 48 texts and got 13 right, and with the lane off that ensemble answered only 38 of 48 (36 correct, 2 wrong) — 8 fewer correct answers bought 2 fewer errors. A strong primary whose band is tight leaves a tie-break lane nothing useful to do, because no AI text defers. Product rule that follows: the lane ships disabled, only a stored model row can enable it, a request body may switch a lane *off* and never *on*, and the admin toggle that used to write only browser `localStorage` now patches the row.
- **Methodological retraction, stated because it changed the numbers above.** An earlier offline audit of this same question hand-copied the band into the audit script and used a hardcoded ruler; it counted 1 deferral of 48 where the shipped rule defers 3, and judged the new primary on the old primary's cuts. Those figures are **retracted**. The audit

was rewritten to execute the shipped evaluator and read thresholds out of the deployed config, and it now refuses to run when the capture's recorded model is not the model whose band is in play. General form: *an offline replay must run the shipped rule, not a transcription of it — a tally without its ruler is unreadable.*

- **The latency operating point is a pool property, and it was repriced rather than re-measured.** Same lane, same day: p50 moved 13.6 s → 96 s between two windows; a full corpus took 144 s once and 5183 s another time, both with zero failures. The lane's ceiling went 90 s → 120 s → 180 s → **240 s**, and the deciding event was a correct answer that arrived too late: a 320-character human text came back HUMAN_LIKELY at p 0.15 with three phrase quotes in **214.2 s** — four seconds past the 210 s the backend was willing to wait, after three timeouts in the preceding half hour. Raising the ceiling buys *coverage*, not a new ruler: the band is untouched, so the accuracy claims hold, and a pool slower than 240 s still answers 502 — now a chosen failure mode with a stated price rather than a bug. The chain must nest: **240 s lane → 270 s caller wait → 285 s WSGI worker → 330 s proxy read**; an outer tier shorter than the inner one converts a slow-but-working detection into an error, which is exactly how that 214 s correct verdict used to be lost.
- **Deploy-time proof (§6's open gap, closed).** Twelve assertions drive the running stack through the proxy on the shipped default lane: health route public vs status route behind login, the served bundle matches the one the repo just built, a signed-in deploy admin, the seeded promoted primary, the execution-purge sidecar attached to the workflow engine's own database, the workflow published with its webhook registered, and two real labelled texts scored end-to-end (browser path → proxy → WSGI → workflow → provider → parser → verdict) with phrase evidence attached and class separation asserted. Measured 2026-10-17: `hu-000` HUMAN_LIKELY p 0.18 in **209.4 s**, `ai-000` AI_GENERATED p 0.80 in **199.5 s**, 3 phrases each — 12/12. The packaged single-user desktop build carries the same lane ceiling and passes 22 of its own assertions, including the seeded `timeout_ms = 240000`.

9.9 Limitations added by Part II

- **The shipped band is capped-era evidence.** 0.80/0.55 was fitted on 2026-10-15 under `max_tokens` and has not been re-validated uncapped; §9.7's paired control says no verdict flipped, but that is a property of the corpus, not proof for the shipping lane.
- **Not cross-validated.** The holdout has a diffusiongemma capture but no `glm-5.3-flash` one, so the tie-break counterfactual in §9.8 is in-sample, and the "lane off" row shares a corpus with the band it is scored at.
- **Stress-corpus provenance is mixed.** Its 33 human rows are from verifiable published sources; its 9 AI rows are self-generated. Treat §9.4 as a hard-case screen, not as an unbiased holdout.
- **Evidence strengths are calibrated to one model.** The 0.30–0.95 domain, the ramp stops and the 99.8% verbatim-locate rate all describe the deployed primary's output habits. Another primary invalidates them.
- **Every timing number here is one gateway's free pool at one moment.** 214.2 s, 209.4 s, 80 s and the 302 s ceiling are pool-regime measurements; a different regime

makes 240 s too small or wasteful, and the application cannot tell which regime it is in except by measuring. Nothing in §9.2–§9.8 is a property of a model.

- **Jitter still bounds what can be claimed.** Borderline verdicts flip on ~8 of 12 texts between identical reruns, and a rerun on the same build minutes apart answered 10/10 at 85.8 s once and timed out the next. Where Part I claimed a structural rather than statistical advantage, Part II's counts (3 wrong of 156, all-3-wrong tie-break) inherit the same limit: they are aggregates, and single-text claims remain unsupportable.
- **Unchanged from Part I:** corpora are English, short-form-heavy, and HC3-derived for the paired measurements; verdicts are style proxies rather than provenance; single-engineer, not peer reviewed.

References

1. [1] Cheng et al., "How Close is ChatGPT to Human Experts? Comparison Corpus, Evaluation and Detection," arXiv:2301.07597 (2022) — the HC3 corpus (six domains of real user answers paired with ChatGPT-3.5 answers).
2. [2] "A Semantic-Invariant Human ChatGPT Comparison Corpus and Its Website," arXiv:2309.02731 (2023) — extended HC3 release used as the datasets server source for row-level provenance checks.
3. [3] McNemar, Q., "Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages," Psychological Methods, vol. 2, 1942. (Exact two-sided binomial form used here.)
4. [4] Mann, H.B., Whitney, D.R., "On a Test of Whether one of Two Random Variables is Stochastically Larger than the Other," Annals of Mathematical Statistics, 1947. (AUC computation: ties at 0.5.)